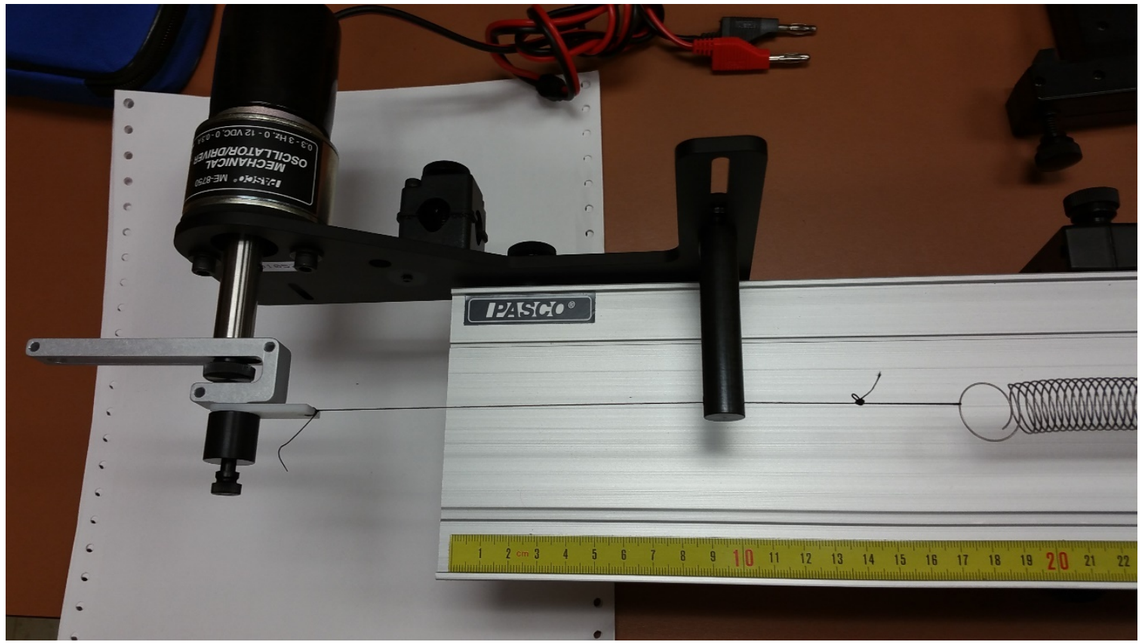
Forced vibration | Lab 5

Repeated motor motion can excite a structural resonance.

Measure impact decay, periodic response and displacement amplification.

MIE 402 · Prof. Ehsan Roohi · Fall 2026

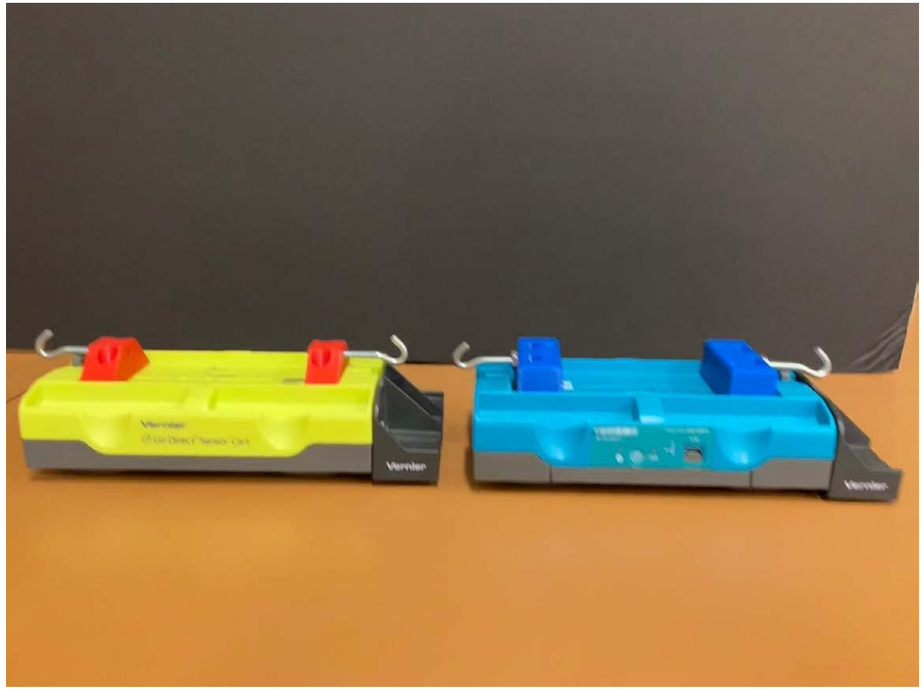


Engineering applications

Pump and motor supports: repeated loading can amplify vibration.

Precision stages: vibration limits positioning accuracy and settling.

This cart test identifies which drive speeds cause large motion.

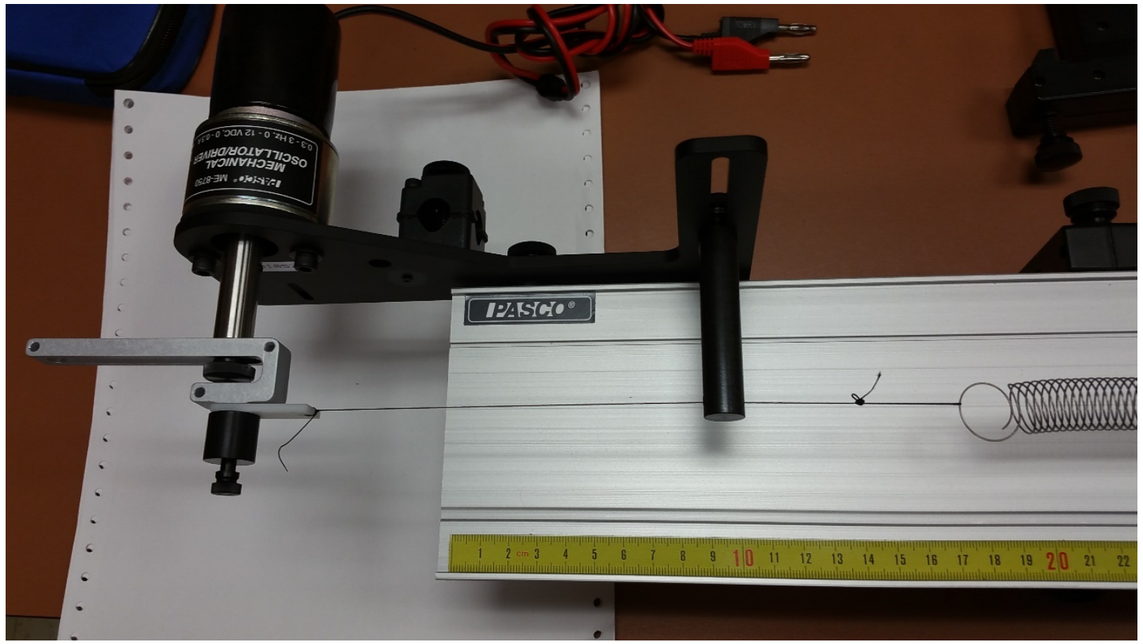


What the apparatus measures

The motor moves one spring end; the cart encoder measures cart position.

Two springs provide restoring force. Magnets provide eddy-current braking.

The tachometer measures drive speed; supply voltage only controls the motor.

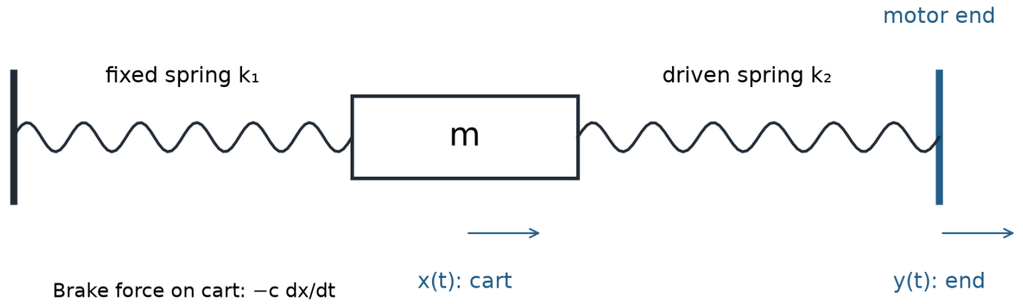


Coordinates and measured parameters

x(t): cart displacement; y(t): driven-end displacement, both positive right [m].

m: total moving mass [kg]; k_1, k_2: spring stiffness [N/m].

c: equivalent braking coefficient [N s/m], acting against the stationary track.

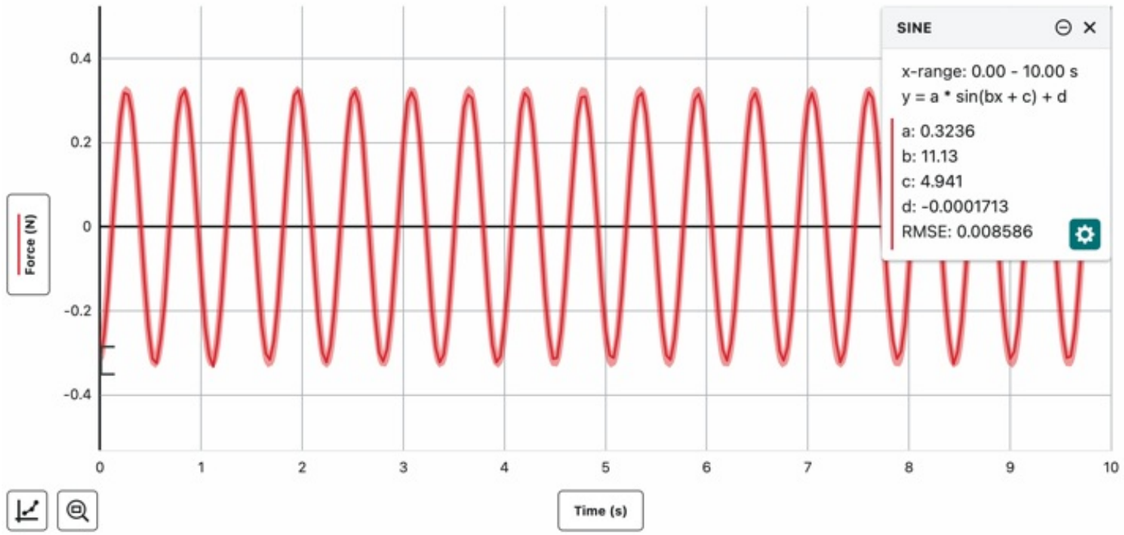


Spring and mass measurements

Weigh the cart, added hex rods and mounted dampers.

For a hanging calibration mass m_h, fit A sin(Bt + phase) + offset.

B is angular frequency [rad/s]; convert Hz before calculating stiffness.



$$k \approx m_h B^2, \qquad B = 2\pi f, \qquad k_{eq} = k_1 + k_2$$

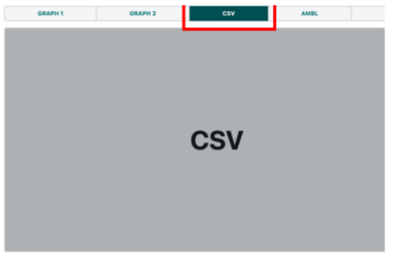
Vernier Graphical Analysis

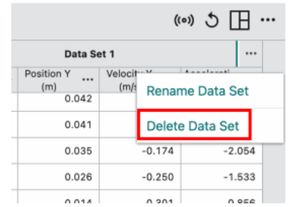
Connect the cart serial number and choose Position versus Time.

Set meters, collection rate at or below 25 Hz, and zero at equilibrium.

Collect before excitation. Export one continuous run as CSV; keep the raw file.







An impulse changes velocity

J: impulse [N s], the time-integral of the applied force F [N].

Velocities v^- and v^+ [m/s] are measured before and after the short impact; position is nearly continuous.

After contact ends, the system moves freely.

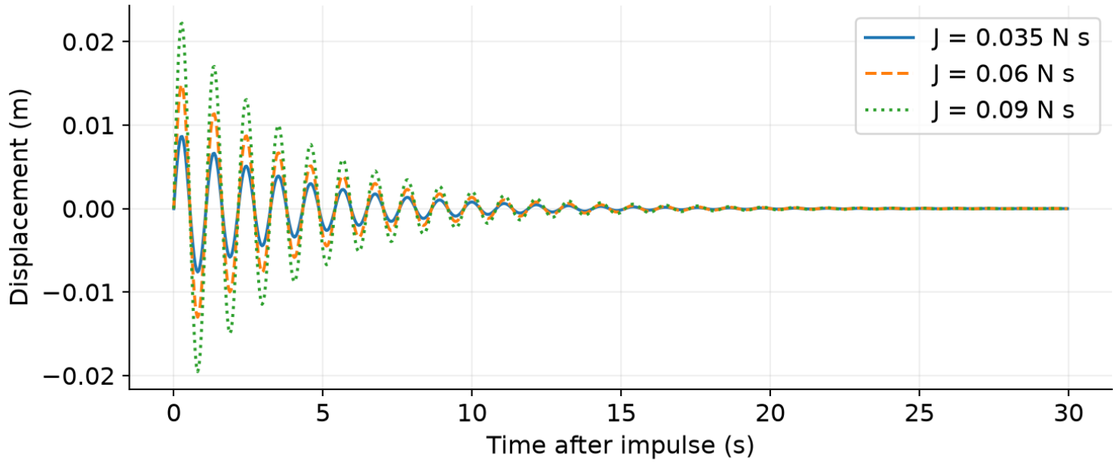
$$J = \int F\,dt, \qquad m(v^+ - v^-) = J$$

Impulse response follows the free-decay model

Define $\alpha = c/(2m)$ [1/s], $\omega_n = \sqrt{(k_e q/m)}$ [rad/s], $\omega d = \sqrt{(\omega_n^2 - \alpha^2)}$.

For a system initially at rest, the impulse sets initial speed J/m.

Three tap strengths test whether amplitude changes while frequency stays similar.



Illustrative model; not laboratory data

$$x(t) = \frac{J}{m\omega_d} e^{-\alpha t} \sin(\omega_d t)$$

What an impulse spectrum tells us

The time record reveals period and decay. The FFT is a frequency cross-check.

A finite decaying record spreads spectral energy; do not infer damping from peak height.

The fitted ωd and model ω_n are distinct; their difference is small for light damping.

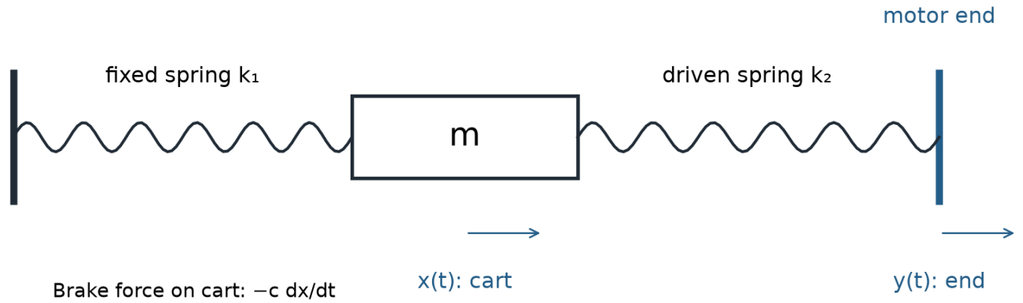
$$f_n = \frac{\omega_n}{2\pi}, \qquad f_d = \frac{\omega_d}{2\pi}$$

The driven spring produces the forcing

The fixed spring pulls by $-k_1x$. The driven spring pulls by $k_2(y - x)$.

The brake opposes cart velocity by $-c\,dx/dt$. Sum these forces.

This gives the laboratory equation directly from Newton’s second law.



$$m\ddot{x} + c\dot{x} + (k_1 + k_2)x = k_2y(t)$$

Steady-state response from the equation

Let y(t) = Y cos(ωt). Y is input peak travel [m]; ω is measured drive frequency [rad/s].

Use complex amplitudes with $i^2 = -1$; differentiation of sinusoidal motion gives iω.

H is the complex displacement ratio. X is cart peak amplitude [m], so |H| = X/Y.

$$H(i\omega) = \frac{\hat{x}}{\hat{y}} = \frac{k_2}{k_1 + k_2 - m\omega^2 + ic\omega}$$

Amplitude and phase of the response

Taking the modulus gives the amplitude ratio. The phase lag ϕ is measured in radians.

The steady output has peak amplitude X and lags the input by ϕ.

Use atan2 so the phase is correct above and below resonance.

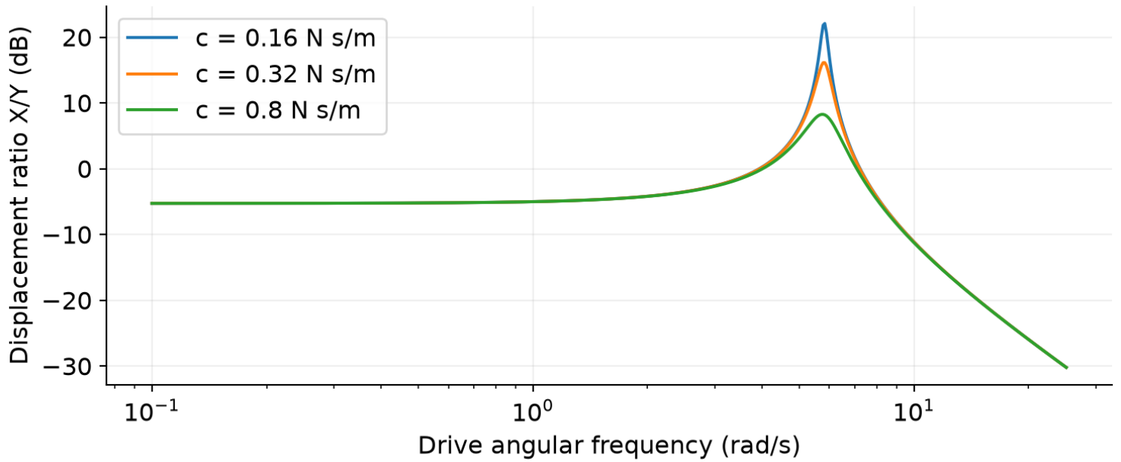
$$\frac{X}{Y} = \frac{k_2}{\sqrt{(k_1 + k_2 - m\omega^2)^2 + (c\omega)^2}}$$

Low-frequency and resonance limits

At low speed the cart follows only part of the endpoint travel: $X/Y \to k_2/(k_1 + k_2)$.

Near the natural frequency, spring and inertia nearly cancel; damping limits amplification.

Above resonance, inertia reduces X/Y. ζ is the dimensionless damping ratio.



Illustrative model; not laboratory data

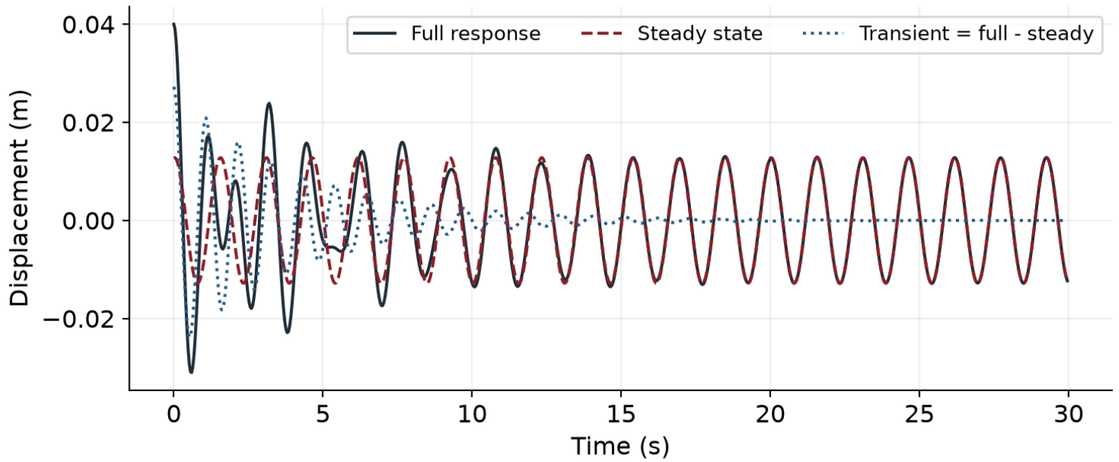
$$\omega_n = \sqrt{\frac{k_1 + k_2}{m}}, \qquad \zeta = \frac{c}{2\sqrt{m(k_1 + k_2)}}$$

The full response includes startup motion

Linearity lets us add the transient and the steady response.

The transient solves the homogeneous equation and decays with exp(−αt).

A settled record follows the drive frequency even if that differs from the natural frequency.



Illustrative model; not laboratory data

$$x(t) = x_{tr}(t) + x_{ss}(t)$$

Transfer function and state equations agree

Laplace transformation with zero initial conditions produces the same H(s).

The state variables are x and v = dx/dt. The second state equation is Newton’s law divided by m.

The pre-lab integrates these equations and checks the steady amplitude against H.

$$\dot{x} = v, \qquad \dot{v} = -\frac{k_1 + k_2}{m}x - \frac{c}{m}v + \frac{k_2}{m}y$$

Measure input travel and motor speed

Measure actual endpoint peak-to-peak travel, typically 2–3 cm, then divide by 2.

Use the tachometer for each run. One revolution corresponds to one drive cycle.

Record the drive amplitude again when the pin setting or endpoint travel changes.

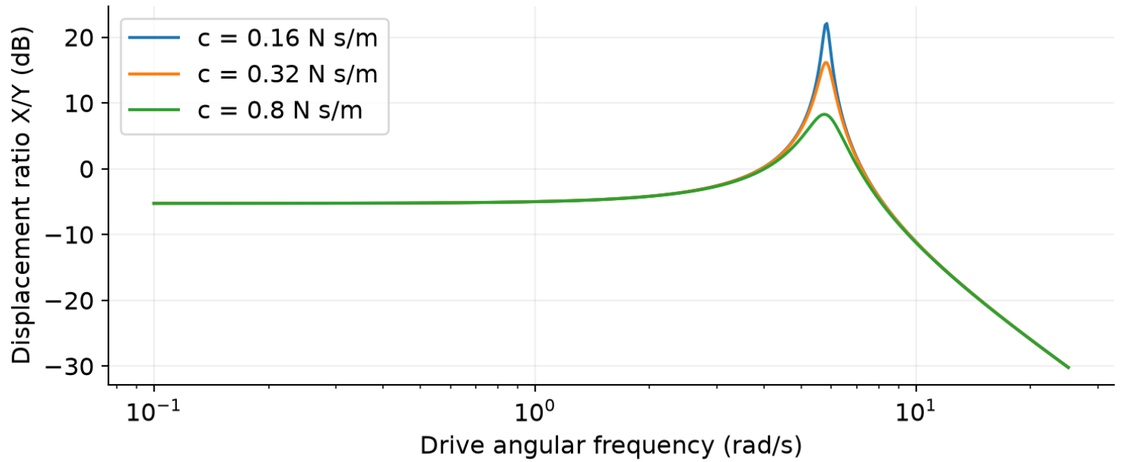
$$Y = \frac{y_{pp}}{2}, \qquad f = \frac{\text{rpm}}{60}, \qquad \omega = 2\pi f$$

A useful sweep covers both sides of resonance

Use 3–5 safe frequencies below and 3–5 above the expected resonance.

Allow startup to decay; record at least five settled cycles, preferably ten.

Keep the springs taut and travel safe. Stop if oscillations become excessive.



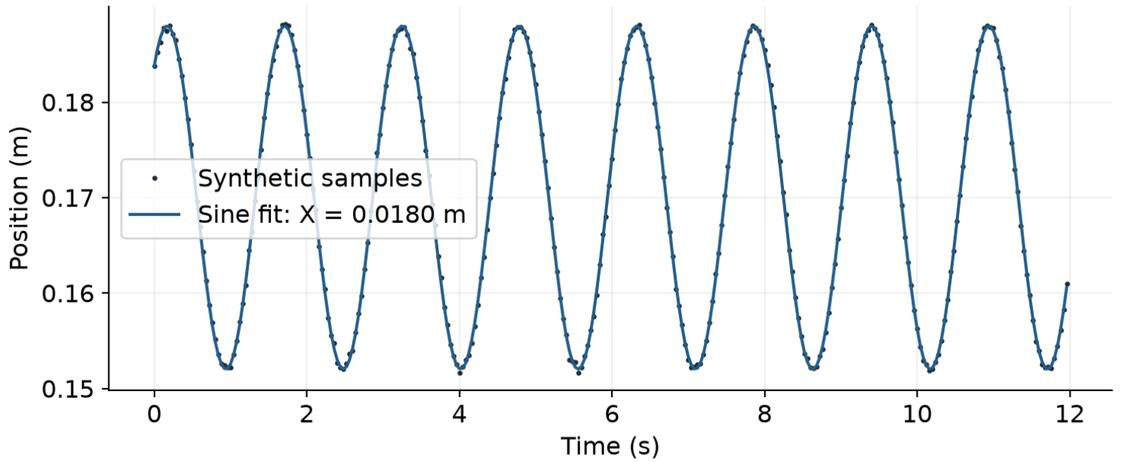
Illustrative prediction used to plan test points

Fit amplitude at the measured drive frequency

Fit x(t) = offset + slow drift + a cos(ωt) + b sin(ωt).

The frequency comes from the tachometer; the code solves for a and b by linear least squares.

The peak amplitude is $X = \sqrt{(a^2 + b^2)}$. Inspect the fit against the data.



Synthetic example of measured-point and fit display

$$G = \frac{X}{Y}, \qquad G_{dB} = 20\log_{10} G$$

Choose damping for the tested configuration

Magnetic dampers change both damping and moving mass.

With the drive off and both ends stationary, record a free decay in the forced-test configuration.

Analyze that decay for c. Use it with the matching mass in the harmonic prediction.

$$c = 2m\alpha, \qquad \omega_d = \sqrt{\frac{k_1 + k_2}{m} - \alpha^2}$$

MATLAB workflow

Unzip the bundle. Open Run_Lab5_Forced_Response.m and click Run.

Choose impulse/decay or harmonic. Enter measured parameters and select CSV or Excel files.

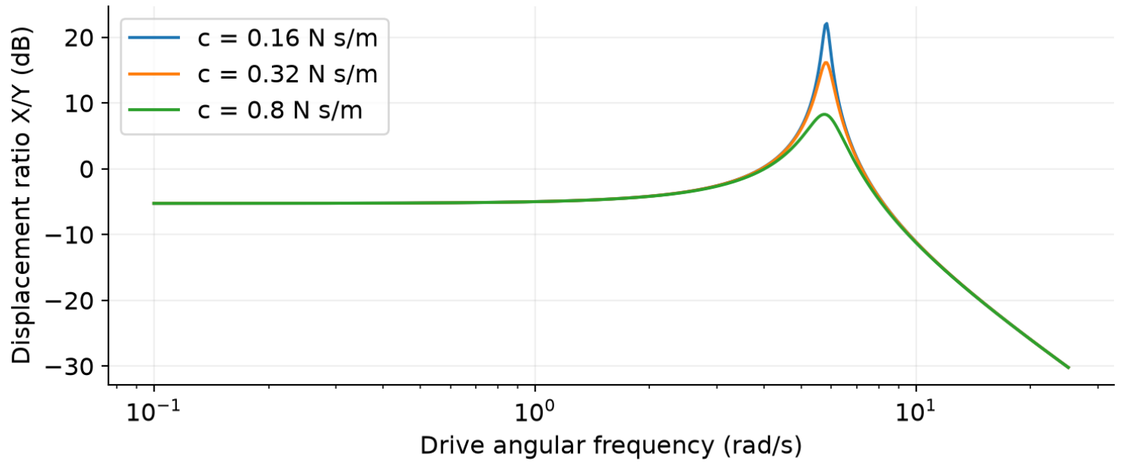
Choose worksheet/columns, a clean time interval and units; harmonic runs also need rpm and input travel.

Figures for the combined Labs 4 + 5 report

Three impulse time histories and spectra: compare frequency and amplitude.

Representative harmonic record: identify startup and the settled interval.

Measured displacement ratios overlaid on theory using your measured parameters.



Illustrative theory only; add your own measured points

Check before leaving the laboratory

Show one impulse trace and spectrum, one settled sine fit and at least five sweep points to the GTA.

Confirm time units, meters, kilograms, peak versus peak-to-peak, and rpm conversion.

Copy the raw files and parameter log to every group member.

Three frequencies to distinguish

Natural frequency: set by mass and the two spring stiffnesses.

Damped transient frequency: appears in decay after release or impact.

Drive frequency: remains in the settled forced oscillation.

$$\omega_n \neq \omega_d \neq \omega \quad \text{in general}$$